

Analyzing passenger train arrival delays with support vector regression



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ABSTRACT

We propose machine learning models that capture the relation between passenger train arrival delays and various characteristics of a railway system. Such models can be used at the tactical level to evaluate effects of various changes in a railway system on train delays. We present the first application of support vector regression in the analysis of train delays and compare its performance with the artificial neural networks which have been commonly used for such problems. Statistical comparison of the two models indicates that the support vector regression outperforms the artificial neural networks. Data for this analysis are collected from Serbian Railways and include expert opinions about the influence of infrastructure along different routes on train arrival delays.

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1. Introduction

Train delays are causing substantial losses to railway operators and passengers. The National Audit Office in United Kingdom reported that there were about 800,000 delays on the British national rail network during 2006–2007. This led to 14 million train-minutes of delays, which cost the passengers about £1 billion in lost time (Burr et al., 2008; Preston et al., 2009). Such numbers provide evidence that reducing train delays in an economically viable way would be highly desirable. An important step in this direction is to establish a functional relation between train delays and various characteristics of a railway system (i.e., infrastructure, timetables and trains). Such a functional relation would allow planners to evaluate how changes in the system would affect delays, and thereby help them determine the changes that would reduce delays in the most economical way.

Let us consider two specific situations in which a functional relation between train delays and characteristics of the railway system would be useful for tactical planning:

- **Investment planning.** Managers are interested in reducing train arrival delays at a station by investing in the infrastructure. The investments may include track renewal to increase the speed profile along some routes, building an overpass to avoid train conflicts, or introducing more advanced signaling systems that resolve rights-of-way more efficiently. A functional relation between train delays and characteristics of the infrastructure would enable the planners to evaluate the effects that different infrastructure projects (or their combination) would have on delays. This would assist the management in selecting the most effective investment plan, given the available budget.

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- **Simulation analysis.** Planners are interested in examining effects of changes in the railway system on bottlenecks, capacity utilization, or delay propagation. Such analysis is typically conducted by simulating specific traffic scenarios, given the information about the infrastructure and train movements (Hansen and Pacht, 2008). Since trains often deviate from their timetables, simulation models must account for delays to validly represent the real-world operations. A functional relation between train delays and characteristics of the system can be used to estimate initial arrival delays to the observed section or intersection. These initial delays can be used further as inputs to a simulation model to obtain more realistic estimates of the capacity utilization, delay propagation, or bottleneck effects at downstream locations.

The described situations motivate research on determining the functional relation between passenger train delays and various characteristics of the railway system. We shall refer to this functional relation as a train delay model that, given a set of inputs (i.e., information about the system), returns train delays. As implied in the motivating examples, we focus on the analysis of historical data and develop train delay models that can be used for tactical planning. This should not be confused with the analysis of train delays at the operational level and development of models for online predictions which are used for real-time traffic management. We review below these two separate lines of research:

- **Operational level.** Models for real-time prediction of train delays are used to detect potential instabilities in the timetable and alert the dispatchers to reschedule some trains. Train delays are predicted from data describing arrivals and departures, which are gathered in real-time Kecman (2014). Train movement data are collected from the infrastructure track occupation records, sensors in rolling-stock, or mobile GPS devices. Flier et al. (2009a) apply an algorithm that sweeps through the delay of the incoming trains and efficiently finds systematic dependencies in large-scale railway delay data. A delay propagation algorithm proposed by Hansen et al. (2010) uses an online model based on timed event graphs to predict running and arrival times. This approach is extended at the microscopic level by Kecman and Goverde (2014), who propose a model based on a timed event graph with dynamic arc weights that accurately captures train behavior and interactions on open track sections and in stations. Berger et al. (2011) propose a stochastic model for delay propagation in large transportation networks, which is suited to process massive streams of real-time data. Delay propagation across the network is modeled with max-plus algebra and analyzed based on timetable properties, such as realizability and stability (Goverde, 2010). A related line of research in the area of traffic predictions develops multivariate state space models that use state variables to describe a system by a set of first-order differential equations. In traffic research, state space models are often applied to multiple input–output systems of stochastic variables (Stathopoulos and Karlaftis, 2003; Dong et al., 2014). State space models for train delays could also process train descriptor messages and historical data to compute train delays using a Kalman filter. An advantage of state space models would be their ability to handle large networks effectively.

As mentioned above, information from online prediction models can be used for real-time traffic management. For example, when delays or disturbances are predicted (or observed), it is vital to quickly retrieve a feasible train schedule. Krasemann (2012) achieves this with a greedy algorithm which efficiently finds good feasible schedules. To minimize train delays and missed connections due to disturbances, Corman et al. (2012) apply two heuristic algorithms to select the connections to be enforced. To reschedule duties of train drivers, Veelenturf et al. (2012) use column generation to solve the crew rescheduling problem with retiming. Corman and Quaglietta (2015) propose a rescheduling tool coupled with a railway traffic simulation model and analyze influence of different rescheduling parameters, prediction models and control delays on system performance. A comprehensive review of the recent literature on rescheduling tools and their applications in train operations is presented by Corman and Meng (2014). The relevant models are classified by their problem scope, model type and solution approach, with a special focus on current train positions and traffic state prediction.

- **Tactical level.** At the tactical level, train delay models have been applied to both timetabling and resource planning. Timetables are tested for robustness using the probability distributions of process durations which are derived from historical traffic realization data. Conclusions from these tests are later used to improve timetable robustness (Medeossi et al., 2011). In a study on passenger trains in the Netherlands, a stochastic approach to cyclic timetable optimization is used to minimize the average delay of trains (Kroon et al., 2008). B ker and Seybold (2012) propose a model with an activity graph where delay propagation is captured analytically, thus enabling precise and efficient calculations even for large networks. Goverde et al. (2013) compare and quantify the infrastructure occupancy both under scheduled and disturbed conditions. Their approach takes into account stochastic delays, dynamic responses to the signaling system, and typical traffic control actions for dealing with conflicts and possibly reducing delay propagation. A comprehensive review of railway disruption management for resource planning (timetable adjustment, rolling stock and crew rescheduling) is presented by Jespersen-Groth et al. (2009).

Various techniques have been used to analyze train delays at both operational and tactical level. They can be roughly classified into three groups:

- **Analytic models.** Probability distributions of train delays are derived from track occupancy and release records, or other data containing information on train movements. The obtained distributions of the underlying random variables are then used within stochastic queuing models (Huisman et al., 2002; Schwanh u er, 1974). An analytic stochastic model is used

to compute delay propagation at platform tracks and junctions (Yuan and Hansen, 2007). For the train arrival delays, different distributions are considered, including Weibull, chi-squared, Erlang and log-normal (Hermann, 1996; Yuan, 2006), but mostly exponential distributions (Higgins and Kozan, 1998; Schwanhäußer, 1974; Yuan, 2006). Huisman and Boucherie (2001) assume exponential distribution for primary delays and forecast secondary delays while accounting for different future traffic scenarios. Goverde et al. (2013) fit a Weibull distribution to empirical data and use it to model initial delays in computing rail line capacity consumption. Meester and Muns (2007) use a phase-type approximation method to address delay propagation in railway networks.

- *Multiple regression.* These models are defined with several variables (i.e., influencing factors) that are correlated with the type of delay that is analyzed. For example, the punctuality of trains in Norwegian railways is correlated with the number of passengers, occupancy ratio, departure punctuality and operational priority rules (Olsson and Haugland, 2004). For the analysis of train dwell times at Dutch railway stations, Wiggeraad (2001) considers the distribution of passengers along the platform, alighting and boarding times, type of station and train service, vehicle characteristics and period of the day. Flier et al. (2009b) predict delays at consecutive stations of Swiss Railways via linear regression and a combinatorial model. This work defines predictors from online train delay monitoring data, such as train density, timetable properties (time between scheduled arrivals, slack time), delays of neighbor trains and previous delays. A statistical analysis of train delays in the Eindhoven Station is used by Goverde (2005) to explain the systematic delay propagation resulting from minor disturbances. The author applies a robust linear regression to investigate the correlation between arrival delays of feeder trains and departure delays of connecting trains. To predict train running times with linear regression, Gorman (2009) considers congestion related delays which are modeled with three groups of input variables: primary (meets, passes, overtaken), secondary (preceding trains, headways, other trains) and capacity utilization factors (train spacing, total train running time).
- *Machine learning.* These models use the presented data to learn a general rule that maps inputs to outputs. They typically provide better fit but less straightforward interpretation than statistical regression models. (Karlaftis and Vlahogianni, 2011) provide a comparison of machine learning and statistical models). Authors mostly use artificial neural networks (ANN) to predict delays of passenger trains (Malavasi and Ricci, 2001; Peters et al., 2005; Yaghini et al., 2011; Yaghini et al., 2013; Pongnumkul et al., 2014). Hybrid models that employ simulation and machine learning (an adaptive neuro-fuzzy inference system) are also used to estimate train delays (Milinković et al., 2013).

Our work is concerned with train arrival delays, which develop due to irregularities in internal sources within the railway system (e.g., infrastructure, train operators, traffic management) as well as exogenous sources (e.g., weather conditions, environment) (Goverde, 2005). Initial models for train arrival delays were analytic. For example, Schwanhäußer (1974) assumed that non-negative arrival delays of passenger trains follow a negative exponential distribution, which was confirmed in later studies (Goverde, 2005). The Weibull distribution is also used to model the non-negative arrival delays (Hermann, 1996). Comparing results based on the Kolmogorov–Smirnov goodness of fit test, Yuan et al. (2006) suggested the use of an exponential distribution as a special case of Weibull for non-negative delays in systems with low traffic density. In terms of their application, arrival delay models were proposed for both operational (Kecman and Goverde, 2014) and tactical (Yuan and Hansen, 2007) planning, and calibrated using detailed data from train describer messages. Arrival delay models are lacking for systems where detailed data on train movements are not available.

This paper proposes machine learning models to analyze train arrival delays on Serbian Railways. We make the following contributions:

- We present the first application of the support vector regression (SVR) in the analysis of train delays and contrast it with ANN, which was successfully used for such problems in the literature. The generalization power of the two models is compared by training and testing them on numerous randomly selected subsets of the data. A statistical comparison is conducted to analyze performance of the models on both data sets. We conclude that the SVR outperforms ANN since it achieves higher average R^2 than ANN on the test data. This result suggests a new direction in applying machine learning models to analyze train delays.
- We explore an expert opinion as an approach for estimating influence of infrastructure on train arrival delays. The influence of multiple factors along a rail line (single-tracking, reduced speeds, characteristics of block and interlocking systems, number of stations, stops, loops, road–rail level crossings and junctions) is aggregated into one variable and its value is determined through the expert opinions of five dispatchers. In the survey, the experts are asked to estimate the possibility of train delays occurring along a particular route due to the aforementioned factors. The final estimates are obtained using the Delphi method, where experts evaluate a route over multiple rounds until a consensus is reached. Data collected from Serbian Railways are used in this paper to show a strong correlation between the expert opinions and train delays.

The remainder of the paper is organized into five sections. First, we discuss train delays on Serbian Railways and analyze input variables. Second, we present two models for predicting delays. Third, we perform statistical comparison to assess predictive power of the models. Forth, we discuss performance of the models and explain potential applications. Fifth, we draw conclusions and suggest extensions of this work.

2. Train delay data

A combination of restricted speed sections, lack of locomotives, and unsuitable timetable design generates train delays and reduces punctuality and reliability on the Serbian railway system. Large disturbances frequently influence the system in such ways that slack times built into the timetables cannot compensate for delays. Since conditions may vary with train routes and trains, it is important to analyze station arrival delays at the level of individual lines and trains. In this regard, arrival delays should be analyzed for each direction separately, as there could be significant differences between trains traveling in opposite directions. This paper studies arrival delays at the Rakovica Station, which is located in the southern part of the Belgrade railway node (Fig. 1). Freight trains that go to the Belgrade marshalling yard from south, west and north do not pass through Rakovica Station, but many of these trains use two neighboring junctions that are controlled by the Rakovica's dispatcher.

The data include 727 passenger trains (99 long-distance, 321 regional, and 307 suburban) going northbound toward Belgrade, and are collected from dispatcher diaries where delays are recorded on a minute scale. Data analysis and interviews with dispatchers reveal several factors that could be influencing arrival delays of passenger trains. An apparent factor is the train category, as higher-category passenger trains have the right-of-way in case of conflicts. Scheduled arrival time should also be considered since we would expect greater delays in peak periods. In addition, we would expect the distance traveled to be strongly correlated with train delays. Intuitively, greater distance implies a higher expected magnitude of train delay. The same argument could be made for time traveled and percent of the journey completed. Finally, the condition of the infrastructure along different lines certainly influences delays. Serbian railways are characterized by: (1) recently renewed lines that enable trains to operate with maximum speed; (2) lines that have sections with temporary speed restrictions; (3) single and double track lines; (4) many junctions and railroad crossings; (5) lines split in sections with different signaling and safety equipment (i.e., block systems: centralized traffic control (CTC), automatic block systems (ABS) and axle counters; mechanical or relay interlocking systems). Consequently, infrastructure conditions vary significantly among different train lines.

To evaluate the influence of infrastructure design and condition on train delays, we have conducted a survey of five experienced dispatchers at the Rakovica Station. They were asked to evaluate each route on a scale from 1 to 10, where 1 denotes a route with the highest number of infrastructural factors that could cause unplanned delays (e.g., a route with many junctions, speed restrictions, long block sections with single-tracking). Infrastructure influence for each route was obtained using the Delphi method (Linstone and Turoff, 1975), where experts answered questionnaires in multiple rounds. They were asked to give their own opinions, after which they were presented with the result summary and encouraged to revise their



Fig. 1. Rakovica station is part of the Belgrade railway node.

earlier answers in order to reach a consensus with other experts. In particular, the survey was used to evaluate infrastructural factors along 39 lines. An example of data presented to the experts is shown in Table 1, which contains information for two train lines. For these two lines, the experts have evaluated the infrastructure influence as follows: Ripanj-Zemun has very low influence (score 9), while Užice – Belgrade has medium influence (score 4). The obtained scores for all the train lines range between 3 and 9.

2.1. Statistics of the variables

Based on the analysis of available data and interviews conducted with dispatchers, we have selected seven factors that might influence passenger train arrival delays on Serbian railways. The seven independent input variables are:

1. Passenger train category (nominal: suburban, regional, long-distance).
2. Scheduled time of arrival at station (continuous).
3. Infrastructure influence defined by expert opinions (ordinal: 3, ..., 9).
4. Percent of journey completed distance-wise (continuous).
5. Distance traveled (continuous).
6. Time traveled (continuous).
7. Headway (continuous).

The correlation between train delays and (continuous and ordinal) input variables is shown in Fig. 2. The scheduled arrival time and headway have statistically insignificant correlation with the response ($r < 0.07$, $p > 0.05$). This is unsurprising as there is a relatively small number of trains and low capacity utilization even in peak hours. Percent of journey completed is moderately correlated with response ($r \approx 0.48$, $p < 0.0001$), while other variables have strong correlations ($|r| > 0.69$) with high significance ($p < 0.0001$). The signs of coefficients in Fig. 2 are also reasonable. For example, distance traveled has a positive coefficient. This is expected, as the chance of having a train delay increases with the distance traveled. A similar argument can be made for time traveled and percent of the journey completed. On the other hand, infrastructure has a negative coefficient, which is intuitive because higher values of that variable indicate better infrastructure conditions along a route, and thus smaller delays. Moreover, the 7×7 correlation matrix is visualized (Komarov, 2013) and also presented in Fig. 2. It shows that scheduled time of arrival and headway are not strongly correlated with any of the other covariates. On the other hand, there are strong correlations ($-0.85 < r < -0.65$) between infrastructure and distance/time traveled or journey completed (denoted by brighter colors in Fig. 2). Percent of journey completed is moderately ($0.55 < r < 0.65$) correlated with distance/time traveled.

We analyzed the distribution of train delays, which typically follows the Gamma, Weibull, or exponential distribution (Yuan, 2006). We fitted 17 continuous parametric distributions (Sheppard, 2012) and ranked them based on the Bayesian information criterion. In Fig. 3 we plotted four distributions that best fitted our data. Additionally, we provided a quantile–quantile (QQ) plot for the exponential distribution that provided the best fit. The QQ plot indicated poor fit of long delays to this theoretical distribution. To improve the fit to theoretical distributions and potentially enhance performance of the models, we considered several transforms to compress the data. However, they neither improved the fit nor performance of the machine learning models. Consequently, the train delay models presented in the following section were calibrated using the original variables.

3. Train delay models

We consider two distinct models to establish a functional relation between train delays and the independent variables. We provide a brief overview of the models and emphasize specifications that yield good performance in analyzing train delays.

Table 1

Data used for expert evaluation of infrastructure influence on train delays.

Line	Ripanj-Zemun	Užice-Belgrade
Stations/Stops/Junctions/Loops/Crossings	3/3/1/1/9	18/16/2/5/29
Percentage of single-track	57	97
Percentage with restricted speed	15	24
Length with restricted speed [km]	1.8	17.9
Block section	BD = 100%	SD = 97%; BD = 3%
Track clear detection	ABS(CTC)=57%; ABS = 43%	AC(CTC)=97%; ABS = 3%
Estimated infrastructure influence [1, ..., 10]	9	4

Note: SD – Station Distance; BD – Braking Distance; ABS – Automatic Block System; CTC – Centralized Traffic Control; AC – Axle Counters;

Variable	r	p -value
Scheduled arrival	0.0694	0.0615
Infrastructure	-0.7106	0.0000
Journey completed	0.4858	0.0000
Distance traveled	0.6947	0.0000
Time traveled	0.7103	0.0000
Headway	0.0027	0.9412

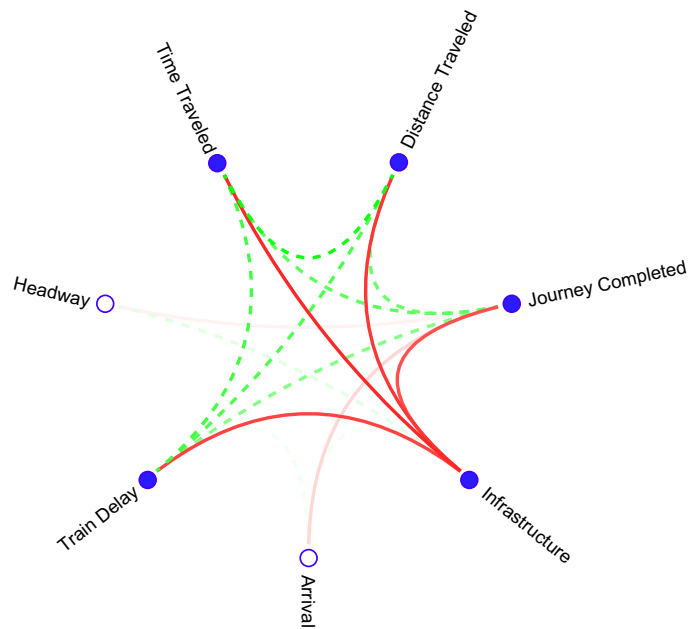


Fig. 2. Numeric values indicate correlation between (continuous and ordinal) input variables and train delays. Green dashed and red solid links denote positive and negative correlation, respectively. Brighter links denote greater correlation, while less visible (or invisible) links imply no correlation. Unfilled tokens indicate that a variable has low total absolute correlation with other covariates (e.g., scheduled arrival time, headway). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

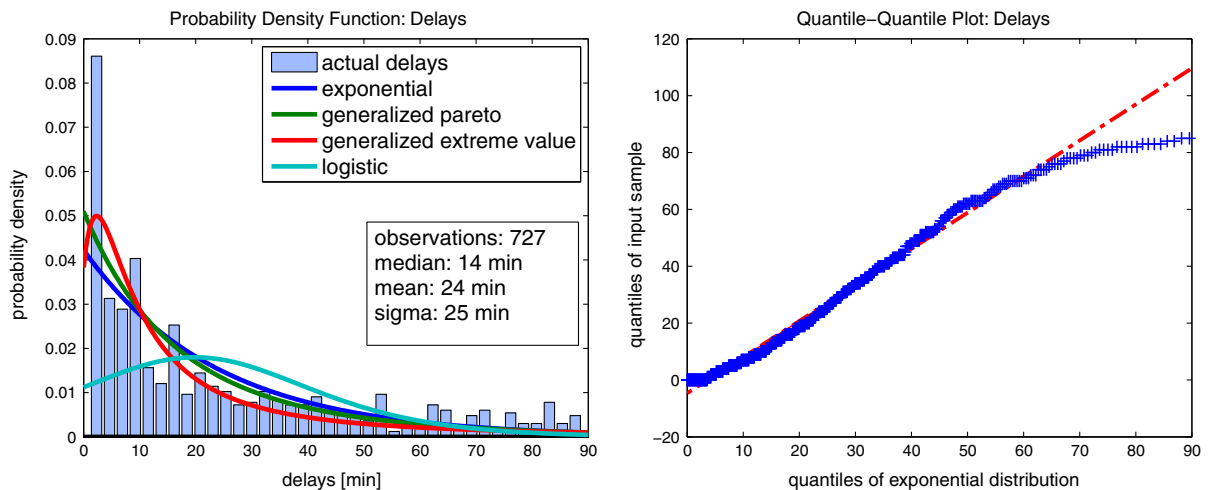


Fig. 3. The exponential distribution fits best the studied train arrival delays.

3.1. Support vector regression

Let $\{(\mathbf{x}_1, y_1), \dots, (\mathbf{x}_n, y_n)\}$ be training data, where $\mathbf{x}_i$ and y_i denote input and target data, respectively (i.e., $\mathbf{x}_i$ is a vector of input variables for the i th train and y_i is its delay). The goal in ϵ -SVR is to determine a function $f(\mathbf{x})$ that (a) has at most ϵ deviation from target data and (b) is as flat as possible (Smola and Schölkopf, 2004). In particular, we apply the ϵ -SVR with radial basis functions

$$f(\mathbf{x}, \mathbf{w}) = \sum_{j=1}^n w_j \exp(-\gamma \|\mathbf{x} - \mathbf{x}_j\|^2), \quad (1)$$

where γ is a parameter and vectors $\mathbf{x}_j$ are inputs from the training data (Broomhead and Lowe, 1988). The vector of unknown parameters $\mathbf{w}$ is determined to minimize the function

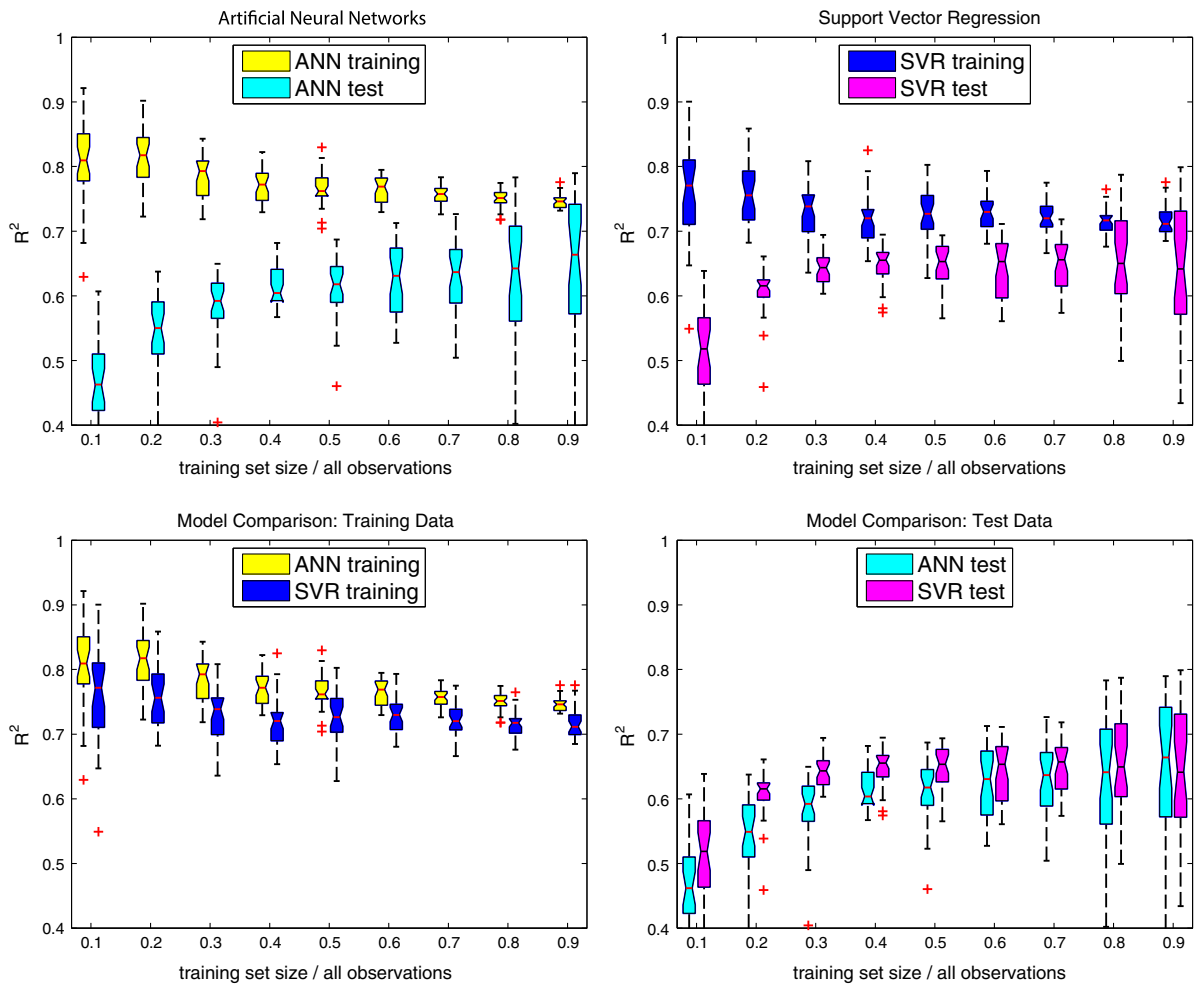


Fig. 4. Comparing R^2 for the ANN and SVR via notched boxplots. The edges of the box are the 25th and 75th percentiles, the whiskers extend to approximately $\pm 2.7\sigma$ and 99.3% coverage, while outliers are plotted individually as red crosses. Box notch indicates 95% confidence interval for the median. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 2

Two-way ANOVA: comparing the ANN and SVR for the training data.

Source of variability	SS	df	MS	F	Prob > F
Models (SVR vs. ANN)	0.2565	1	0.2565	182.0885	0.0000
Ratio (0.1–0.9)	0.2092	8	0.0261	18.5622	0.0000
Interaction between models and ratio	0.0138	8	0.0017	1.2244	0.2823
Error	0.7352	522	0.0014	–	–
Total	1.2146	539	–	–	–

Note: SS – sum of squares; df – degrees of freedom; MS – mean squares (SS/df); F – F statistics (ratio of MS).

Table 3

Two-way ANOVA: comparing the ANN and SVR for the test data.

Source of variability	SS	df	MS	F	Prob > F
Models (SVR vs. ANN)	0.2365	1	0.2365	39.2234	0.0000
Ratio (0.1–0.9)	1.5986	8	0.1998	33.1417	0.0000
Interaction between models and ratio	0.0898	8	0.0112	1.8626	0.0636
Error	3.1474	522	0.0060	–	–
Total	5.0724	539	–	–	–

Table 4
Mean R^2 based on all test instances.

	ANN	SVR
95% confidence interval (training data)	[0.76938, 0.77896]	[0.72528, 0.73590]
95% confidence interval (test data)	[0.57141, 0.59888]	[0.61866, 0.63534]

Note: The SVR statistically outperforms ANN for the test data. The average R^2 for SVR (0.627) is 7% higher than for ANN (0.585).

$$\min_{\mathbf{w} \in \mathbb{R}} \frac{1}{2} \|\mathbf{w}\|^2 + C \cdot \sum_{i=1}^n \max(|y_i - f(\mathbf{x}_i, \mathbf{w})| - \epsilon, 0), \quad (2)$$

where parameter $C > 0$ controls the tradeoff between the flatness of $f(\cdot)$ and amount up to which deviations greater than ϵ are tolerated. The dual of this optimization problem is solved using convex programming techniques (Smola and Schölkopf, 2004).

Data preprocessing and parameter selection are important for good SVR performance (Hsu et al., 2003). First, all the variables are scaled linearly to the range $[0, 1]$. Second, our ϵ -SVR includes parameters γ , C , and ϵ . We wish to find the values of these parameters that maximize the predictive power of our model. This is done via combinations of cross-validation and grid-search techniques. We use the 5-fold cross-validation (Hsu et al., 2003) and perform a simple grid search over the following range of parameters: $\gamma = 2^a$, $a \in \{2, \dots, 8\}$; $\epsilon = 2^b$, $b \in \{-4, \dots, 4\}$; $C = 1.1^c$ ($\max_i y_i - \min_i y_i$), $c \in \{-22, \dots, -12\}$.

3.2. Artificial neural network

ANN models combine elements of learning and adaptation (Palit and Popović, 2005). An ANN is trained to perform a particular function by adjusting the values of the connections (weights) between elements. The difference between the output and the target is sent to the learning procedure, which keeps readjusting the weights until the network output matches the target sufficiently. A common training procedure is the backpropagation algorithm (Palit and Popović, 2005) that adjusts the network weights and biases in the direction in which the error performance function decreases most rapidly. The topology of the network is usually specified with the number of neurons and neural layers.

The ANN was trained using the independent variables from Section 2.1 as input data, and train delays as the corresponding target data. The categorical variables were previously converted to groups of binary variables. We used a soft computing function approximator: a hidden 50 neuron layer and an output map consisting of feed forward net with tangent hyperbolic transfer function. The ANN was trained with the Levenberg–Marquardt backpropagation algorithm. Before training, input and target data were randomly divided into three sets which are used (a) for training, (b) to validate that the network is generalizing and to stop training before overfitting (Kecman, 2001), and (c) to test the generalization power of the network. As an additional approach to avoid overfitting, we trained 100 independent ANN models and then averaged the outputs they provide.

4. Results

All models were implemented in MATLAB. The ANN is included as part of the statistical toolbox. The ϵ -SVR is implemented using LIBSVM (Chang and Lin, 2011), which is a popular open source library for support vector machines.

4.1. Model comparison

To compare SVR and ANN, a subset of the total data set is randomly selected to train the models, while the remainder is used to test their generalization power. Models are tested on instances where the sample size used for training ranges from 10% to 90% of the total sample. For each ratio, models are tested on 30 runs with randomly selected observations. The corresponding results are summarized in Fig. 4, which presents boxplots for the computed R^2 (i.e., coefficient of determination). The first row in Fig. 4 shows the effect of overfitting to noise (i.e., high R^2 for training and low for test data). It indicates that the ANN exhibits greater discrepancy in the R^2 for training and test data. The second row in Fig. 4 compares performance of the two models for training and test data, respectively. To statistically compare their performance, we do the following:

- We conduct a two-way analysis of variance (ANOVA) to test for the difference in mean R^2 . For training data, there is a difference between the models (Table 2). For the test data, there is also a difference in the performance of ANN and SVR (see Table 3).
- We provide 95% confidence intervals for mean R^2 based on all test instances (Table 4). The confidence intervals reiterate conclusions from ANOVA, and allow us to rank the models. For training data, the ANN was better than the SVR. However, these results are less relevant because they can be due to overfitting. More importantly, for the test data, the SVR outperforms the ANN.

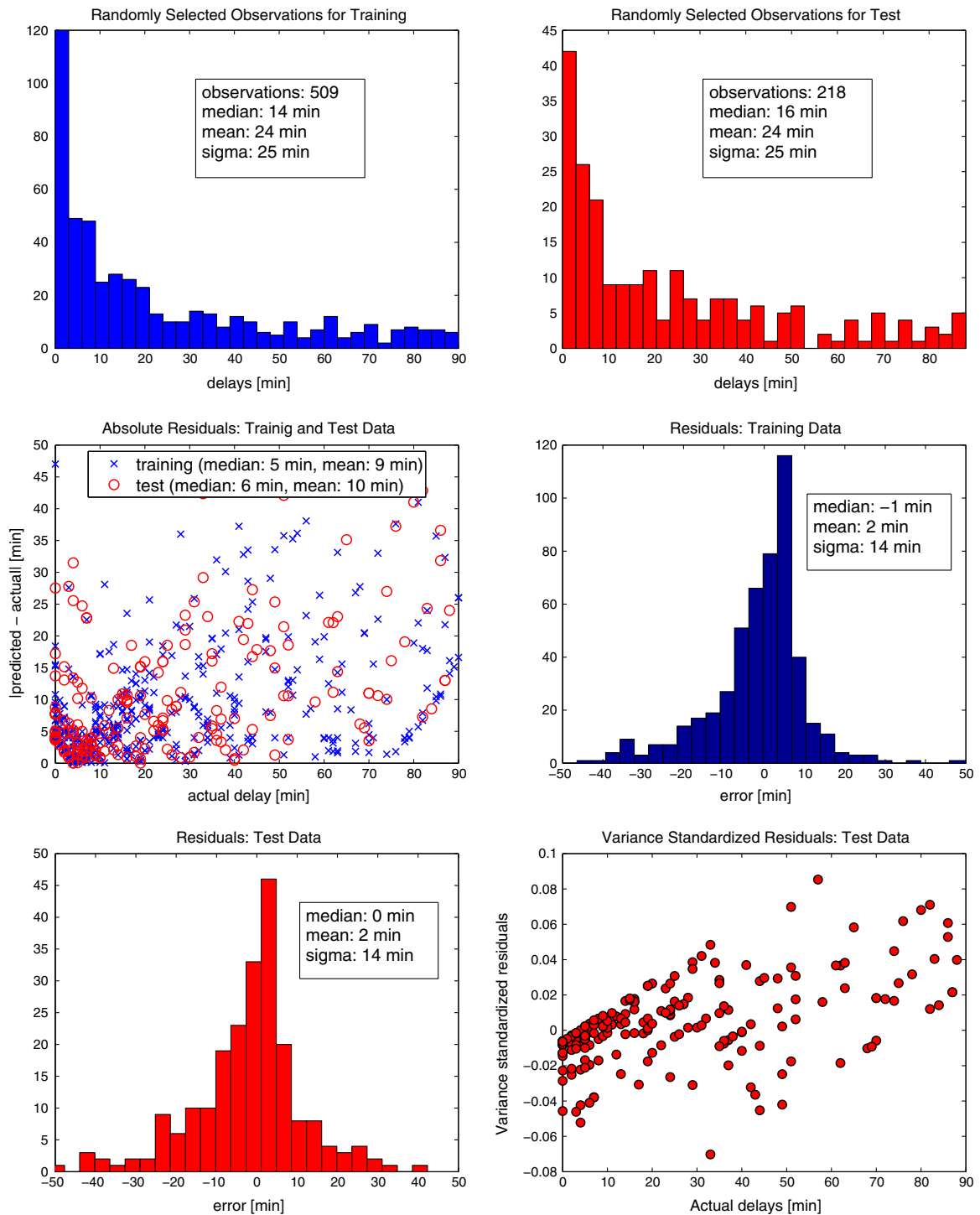


Fig. 5. An example application of SVR on two randomly selected subsets for training and test.

4.2. SVR example application

We saw that statistical comparison indicated better generalization power for the SVR over the ANN. Here we explore performance of the SVR on a concrete instance, which includes 509 randomly selected observations for training and the remaining 218 for testing, as shown in the summary statistics in the first row in Fig. 5. The obtained R^2 values for training and test data are 0.69 and 0.68, respectively. The following three plots in Fig. 5 show absolute residuals and histogram plots of the

raw residuals. The computed median absolute error is 5 and 6 min for training and test data, respectively (i.e., 35.7% and 37.5% of the median delays in two samples). The last plot in Fig. 5 shows variance standardized residuals for the test data. It indicates relatively high errors for very short delays (i.e., below 5 min). This can be attributed to the practice of sometimes registering trains with slight delays as if they arrived according to the timetable, which adds noise to very short delays. Also, recording arrival times in minutes adds more noise to short delays than to long ones.

5. Discussion

We have presented the first application of the SVR in the analysis of train delays and showed that it achieves better performance than ANN. Since ANN models are commonly used to study train delays (Malavasi and Ricci, 2001; Peters et al., 2005; Yaghini et al., 2011; Yaghini et al., 2013; Pongnumkul et al., 2014) and they generally outperform the widely applied multiple regression (Wiggenraad, 2001; Olsson and Haugland, 2004; Goverde, 2005; Gorman, 2009; Flier et al., 2009b), we believe that our results point out a new direction in this line of research. One of the main characteristics of SVR is that instead of minimizing the training error like ANN, SVR attempts to minimize a weighted sum of a regularization term and training error (Eq. (2)) in order to achieve a generalized performance. Indeed, our results show that ANN outperforms SVR on the training data. However, the more relevant comparison of the two models on the test data indicates better generalization power for the SVR.

We have already given generic examples of how a functional relation between train delays and system characteristics could be used in practice. Here, we reiterate potential applications that stem from the variables considered in our work. In the following examples, we reference variables according to their numbering from Section 2.1.

- Consider the case when a problem with the infrastructure is detected, which would require the operator to introduce a temporary speed restriction until the problem is resolved. This would increase travel times of trains traversing the affected section (variable #6), postpone their scheduled arrival times (variable #2), and change headways for all trains (variable #7) arriving at a station. The proposed SVR model could be used to estimate train arrival delays for these new conditions. Estimating effects of future speed restrictions would enable the operator to immediately determine the priority (or a time window) for resolving the aforementioned problem with the infrastructure. In this regard, it should be noted that maintenance crews are often assigned time windows for their various tasks, which can depend both on traffic conditions and priorities of projects.
- Consider the renewal of a double-track railway which is sometimes done by closing a section of a track (which constitutes a “work zone”) in one direction and diverting the trains to the other track. This introduces single-tracking, which is considered in determining the influence of the infrastructure (variable #3) on train delays. (Recall that experts, among other factors, look at the percentage of a route with single-tracking, Table 1.) Closing longer sections of track, i.e. lengthening work zones, reduces maintenance cost per kilometer due to (a) economies of scale and (b) smaller fixed cost associated with repeatedly setting up work zones. However, it increases delay cost due to single-tracking. The proposed SVR model could be used to estimate train delays for different lengths of work zones, and help determine which one is preferable. Note that this application would require experts to estimate the influence of the infrastructure, given different percentages of a line with single-tracking.

6. Conclusions

This paper proposed machine learning models for analyzing the relation between train arrival delays and various characteristics of the railway system, which is important for planning changes and investments aimed to reduce delays. We presented the first application of SVR in the analysis of train delays. The performance of the model was compared with an ANN, which was commonly used for similar problems in the literature. The two models were trained and tested on numerous randomly selected subsets of the data collected from Serbian Railways. Statistical comparison of their generalization power indicated better performance of the SVR. This result encourages further applications of the SVR in the analysis of train delays.

The proposed SVR can be applied to analyze train arrival delays at other stations, provided that similar types of data are collected to calibrate the models. The current methodology accounts for characteristics of the infrastructure design and condition via expert opinions. An interesting extension would be to capture these influences through input variables that might be quantified without the need to interview experts. These variables would include the number of junctions and stations, type of signalization, and other characteristics of the infrastructure. Since this extension would include far more variables, it would require extensive data collection to calibrate the train delay models. Another possible extension would be further stratification of data because short and very long delays can be associated with different factors (e.g., long delays are often due to incidents which are not captured with the variables considered in this study). This approach would also require a larger data sample.

As argued earlier, current models can be used to evaluate effects of different infrastructure projects on delays, and thereby help planners compare different investment alternatives and select the most favorable ones. An interesting extension may include development of an optimization approach to select a near-optimal investment plan, given the available budget. The proposed train delay models can be used to evaluate the objective function (i.e., costs associated with train delays) and,

thereby, direct the algorithmic search towards a near-optimal investment plan. Such an approach could be developed to optimize investment decisions over the course of multiple years, assuming different sources of stochasticity (e.g., uncertain variation in train traffic intensity, budgets and investment costs).

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